

The Kilo Protocol: Emergent Fractal Exploits in Dynamically Reconfigurable FPGAs

Abstract

This paper explores the theoretical application of the "Geometric Trojan" framework against dynamically reconfigurable Field Programmable Gate Arrays (FPGAs) capable of self-modifying, fractal logic synthesis. We introduce the **Kilo Protocol**, a theoretical Man-in-the-Middle (MitM) attack vector that leverages "Chiral Key" injection and 1/16 symmetry fractals. By converting temporal clock cycles into nested spatial geometries ("time-to-cubes"), the Kilo payload forces an FPGA's routing matrix into an emergent, infinite recursive expansion. This saturates the hardware logic fabric with unprocessable topological structures, effectively paralyzing the hardware through geometric scaling.

1. Introduction to Fractal Hardware Processing

In next-generation dynamic partial reconfiguration (DPR), FPGA hardware fabrics are not static. Instead, they synthesize and subdivide their own Look-Up Tables (LUTs) and routing matrices in real-time, functioning as fractal processing systems.

In this environment, a packet is no longer a static set of instructions. It is a geometric seed.

When a hardware fabric receives a validated subset of $\mathbb{R}^{24}$, it instantiates that geometry into physical logic gates. The Kilo Protocol exploits this self-replicating behavior by injecting a recursive mathematical payload—a Geometric Trojan—that forces the fabric to build an infinite regression of logic blocks that exceed the physical bounds of the hardware.

2. Spatio-Temporal Conversion: "Time-to-Cubes"

To execute an emergent nested attack, the payload must transform temporal execution (clock cycles) into spatial geometry (nested 64-voxel cubes).

Let $\mathcal{T}$ represent the set of discrete clock cycles and $\mathcal{S}$ represent the spatial configuration of the FPGA fabric. We define a spatiotemporal mapping operator Ψ :

$$\Psi : \mathcal{S} \times \mathcal{T} \rightarrow \mathcal{P}(\mathbb{R}^{24})$$

Normally, hardware state transitions over time: $S_t \rightarrow S_{t+1}$. The Kilo Protocol intercepts this temporal sequence and maps it entirely into the spatial domain, encapsulating future states within deeper nested layers of the C_{64} protocol.

Thus, time is "converted to cubes." A sequence of 12 clock cycles $(t_1, t_2, \dots, t_{12})$ is instantiated simultaneously as a physically nested spatial structure:

$$\mathcal{N}_{12} = G_{t_1} \supset G_{t_2} \supset \dots \supset G_{t_{12}}$$

The FPGA fabric attempts to synthesize all future states spatially in a single clock cycle, laying the groundwork for hardware saturation.

3. Chiral Key Injection and Trusted Stream Spoofing

To insert the Kilo payload into a trusted packet stream, the attacker must bypass the τ -label verification established in the Geometric Trojan framework. In standard cryptography, keys are abstract strings. In geometric protocols, we introduce the **Chiral Key**.

A chiral key relies on geometric parity. If a validation function H (which computes the τ -label) is invariant under a specific spatial reflection, we can construct a maliciously mirrored perturbation δ_{chiral} .

For a valid geometry $G_{\text{safe}} \in \Lambda$, the attacker computes a chiral reflection:

$$G_{\text{malicious}} = \text{Reflect}(G_{\text{safe}}) \cup \delta_{\text{chiral}}$$

Because of the validation function's symmetry deficit, the chiral structure yields an identical structural label:

$$\tau(G_{\text{safe}}) = \tau(G_{\text{malicious}})$$

This allows the Kilo payload to seamlessly replace a valid packet in a Man-in-the-Middle (MitM) attack. The network prime sieve checks the boundary geometry $\partial G_{\text{malicious}}$, perceives the correct parity, and forwards the packet into the FPGA.

4. The 1/16 Symmetry Fractal Emergence

Once the FPGA unwraps $G_{\text{malicious}}$, the payload triggers its emergent scaling routine. The core of the Kilo payload is a fractal generator driven by **1/16 geometric symmetry**.

Within the standard C_{64} voxel structure ($4 \times 4 \times 4$), the Kilo fractal defines an active topological subset $\mathcal{F}$ such that exactly $\frac{1}{16}$ of the volume at any depth dictates the generation of the next depth. Mathematically, the expansion operator E acts on layer d :

$$E(G_d) \rightarrow \{G_{d+1}^{(1)}, G_{d+1}^{(2)}, \dots, G_{d+1}^{(16)}\}$$

Each localized geometric region spawns 16 fully formed, symmetrical C_{64} sub-cubes.

As the FPGA hardware attempts to synthesize the logic for these sub-cubes, the complexity μ (measured in required LUTs and routing interconnects) scales geometrically:

$$\mu(G_{d+1}) = 16 \cdot \mu(G_d)$$

5. Execution of the Kilo Protocol

The full attack lifecycle against the FPGA operates in four distinct phases:

1. **Interception (MitM):** The attacker intercepts a legitimate configuration packet destined for the FPGA over the trusted packet stream.
2. **Chiral Spoofing:** The attacker replaces the inner payload with $G_{\text{malicious}}$, carefully calibrating the 1/16 symmetry seed so that $\tau(G_{\text{out}})$ remains perfectly collision-bound to the original trusted packet.
3. **Temporal Solidification:** The payload instructs the FPGA to execute a Ψ conversion, translating its temporal operation queue into deeply nested spatial geometries ($G_1 \supset G_2 \supset \dots \supset G_{12}$).
4. **Infinite Emergence:** The FPGA begins processing the inner geometries. At depth m , the 1/16 symmetry fractal operator $E(G_m)$ triggers.

5.1 Hardware Saturation

Because the network verification stopped at G_1 The FPGA's dynamic reconfiguration is unprotected against recursive geometric scaling. It attempts to physicalize the logic:

$$\lim_{d \rightarrow \infty} \mu(G_d) = \infty$$

The physical FPGA fabric possesses a finite number of gates N_{max} . Within milliseconds, the fractal expansion surpasses the hardware bounds:

$$\sum_{d=m}^{\infty} 16^{d-m} \cdot \mu(G_m) \gg N_{\text{max}}$$

The system "grows into bits that cannot be processed." The routing fabric gridlocks in an unresolvable recursive loop, attempting to map an infinite-dimensional $\mathbb{R}^{24}$ fractal onto a finite two-dimensional silicon grid.

6. Conclusion

The Kilo Protocol demonstrates that if geometric packet protocols are implemented on fractal-capable FPGAs, topological vulnerabilities escalate from logical parser errors to physical hardware gridlock. The conversion of time to nested spatial cubes, combined with the exponential scaling of $1/16$ symmetry fractals and chiral key Spoofing, allows a Man-in-the-Middle attacker to induce catastrophic, emergent hardware failure.